

PyGameMaker — Tutorial 12: 2.5D Textures, Sky and Floor — Teacher Guide

Companion for the in-app tutorial (**Help > Tutorials > 2.5D: Textures, Sky and Floor**, 4 pages) and the matching student handout and worksheet. Students continue their project from Tutorial 11. The 2.5D lessons are **hidden in the beginner edition**: switch the edition first.

Overview

Students dress up the first-person room with textures. New ideas: a **texture is just a sprite** chosen in *Enable Raycast View*, a **panning sky**, a **tiled floor**, **fallback colours**, and the **detail versus speed** settings. The game does not change, so this is a good art and design lesson.

Suggested Timing (45 minutes)

Segment	Time	What happens
Recap and predict	5 min	Show flat grey walls vs textured; ask what makes it look real
Phase 1: textured walls	10-15 min	Pages 1-2; drawing the brick takes time, or import an image
Phase 2: sky and floor	15 min	Page 3
Phase 3: tune the look	10 min	Page 4
Worksheet / exit ticket	5 min	Worksheet parts A-C

Step-by-Step Walkthrough and Common Issues

Phase 1: textured walls

- A texture is an ordinary sprite. Set **Wall Texture** on *Enable Raycast View* to the sprite's exact name. **Square** pictures such as 64×64 look best.
- If **Wall Texture** is empty the walls are drawn in the flat **Wall Color**. **Textured Walls = off** forces flat colour even when a texture is set (useful for comparing).
- Walls facing away from the light are drawn darker; this is a normal shading trick that makes corners look solid.

Phase 2: sky and floor

- The **sky** is a wide picture (256×64). It pans sideways as you turn (one full turn slides the picture past once) but does not change size as you walk, like a far-away horizon.
- The **floor** tile should be 32×32, one grid cell. It repeats once per cell and lines up with the bottoms of the walls.
- **Ceiling Texture** is used **only when no sky is set**; a Sky Texture always wins.
- Empty names mean fallback colours (Wall, Floor and Ceiling Color). In the reference project, clearing the texture names gives the flat colours again.

Phase 3: tune the look

- **Columns** (default 320) and **Floor Detail** (default 4) trade detail for speed: fewer columns and a higher Floor Detail are faster but chunkier. A good order to try on a slow computer: raise Floor Detail to 6-8 first, then lower Columns.

- Typos in texture names are the most common problem; ask students to copy names.

Reference projects. Download the checkpoint projects from the wiki (12_raycast_textures_checkpoints.zip): the project at the start of the lesson (the result of Tutorial 11) and the finished, textured project. The textures in the reference project are simple generated images; students should draw or import their own.

Vocabulary Introduced

Term	What it means here
Texture	A sprite used to cover a wall, floor or ceiling
Panorama	A wide picture that pans as you turn (the sky)
Tile	A small picture repeated across the floor
Fallback colour	The plain colour used when a texture is not set
Columns / Floor Detail	The two speed versus detail settings

Discussion Questions

- Why does the sky move when we turn but not when we walk?
- Why should a floor tile match the size of a grid cell?
- How do old games get away with such simple pictures?
- What would you change to make a spooky level?

Differentiation

- **Support:** provide ready-made texture images to import; start from the finished project and change one texture.
- **Extension:** several wall looks; an indoor ceiling; a cave with dark colours and short Render Distance.

Worksheet Answer Key

Part A: 1-C, 2-D, 3-A, 4-B. **Part B:** 1. Wall Texture. 2. Sky Texture. 3. Floor Texture. 4. Set Textured Walls off. 5. Raise Floor Detail. **Part C:**

- 1 The sky: a Sky Texture always wins over a Ceiling Texture.
- 2 The floor is drawn one tile per cell, so a 32×32 tile lines up with the cells and the wall bottoms, and the floor does not seem to slide.
- 3 That the texture name is spelled exactly like the sprite, and that **Textured Walls** is on.

Parts D and E: completion and reflection.

Rubric: 2.5D Textures

Level	What the project shows
4 - Complete	Textured walls, a sky that pans and a tiled floor; the student can explain the fallback colours and the detail settings
3 - Working	Walls and either the sky or the floor are textured

2 - Partly there	Only the walls are textured, or names are wrong so some textures do not show
1 - Started	The room still shows flat colours

End-of-Session Checklist

- ☐ Every student has a saved project (Ctrl+S)
- ☐ Every student saw a textured wall
- ☐ Note who needs the checkpoint project next time